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soumith committed on Jun 6, 2017

```
fix incorrect grad_weight in Bilinear
```

main (#1739) · whc_flight_4 bc2caa7fdf006894eff7af936babde69ab5a40f8-huydhn-debug

1 parent b6c75c4 commit d50ad40

2 files changed +20 -12 lines changed

Search within code

+17 -10

test/test_nn.py

@@ -2130,29 +2130,36 @@ def test_cosine_similarity(self):

```
2130  2130
2131  2131          def test_bilinear(self):
2132  2132              module = nn.Bilinear(10, 10, 8)
2133        -         module2 = legacy.Bilinear(10, 10, 8)
      2133  +         module_legacy = legacy.Bilinear(10, 10, 8)
2134  2134
2135        -         module2.weight.copy_(module.weight.data)
2136        -         module2.bias.copy_(module.bias.data)
      2135  +         module_legacy.weight.copy_(
```

```
                module.weight.data)
2136            +
                module_legacy.bias.copy_(mo
                dule.bias.data)
2137 2137
2138 2138              input1 =
                torch.randn(4, 10)
2139 2139              input2 =
                torch.randn(4, 10)
2140 2140
2141 2141              output =
                module(Variable(input1),
                Variable(input2))
2142            -         output2 =
                module2.forward([input1,
                input2])
2142            +         output_legacy =
                module_legacy.forward([inpu
                t1, input2])
2143            +
2144            +
                self.assertEqual(output.dat
                a, output_legacy)
```

```
2143  2145
2144  2146             input1_1 =
                       Variable(input1,
                       requires_grad=True)
2145  2147             input2_1 =
                       Variable(input2,
                       requires_grad=True)
2146  2148
2147        -          output3 =
                       module(input1_1, input2_1)
2148        -          grad =
                       torch.randn(*output3.size()
                       )
2149        -
                       output3.backward(grad)
      2149  +          module.zero_grad()
      2150  +
                       module_legacy.zeroGradParam
                       eters()
      2151  +
      2152  +          output =
                       module(input1_1, input2_1)
      2153  +
```

```
                grad_output =
torch.randn(*output.size())
2154 +          gi1_legacy,
gi2_legacy =
module_legacy.backward([inp
ut1, input2], grad_output)
2155 +
output.backward(grad_output
)
2150 2156         gi1 =
input1_1.grad.data.clone()
2151 2157         gi2 =
input2_1.grad.data.clone()
2152 2158
2153 -
self.assertEqual(output.dat
a, output2)
2154 -           # TODO: this
assertion is incorrect, fix
needed
2155 -           #
self.assertEqual([gi1,
gi2], output3)
```

```
2159 +
       self.assertEqual(gi1,
       gi1_legacy)
2160 +
       self.assertEqual(gi2,
       gi2_legacy)
2161 +
       self.assertEqual(module.wei
       ght.grad.data,
       module_legacy.gradWeight)
2162 +
       self.assertEqual(module.bia
       s.grad.data,
       module_legacy.gradBias)
2156 2163
2157 2164
       self.assertTrue(gradcheck(l
       ambda x1, x2:
       F.bilinear(x1, x2,
       module.weight,
       module.bias), (input1_1,
       input2_1)))
2158 2165
```

+3 -2

torch/nn/_functions/linear.py

```
@@ -73,13 +73,14 @@ def backward(self, grad_output):
73  73              buff.mul_(grad_output.narrow(1, k,
                    1).expand(grad_input2.size()))
74  74              grad_input2.add_(buff)
75  75
    76  +           grad_weight = weight.new(weight.size())
76  77              if self.needs_input_grad[2]:
77  78                  # accumulate parameter gradients:
78  79                  for k in range(weight.size(0)):
79  80                      torch.mul(input1,
```

```
                grad_output.narrow(1, k,
                1).expand_as(input1),
                out=buff)
80              -               grad_weight =
                torch.mm(buff.t(), input2)
                81              +               grad_weight[k]
                = torch.mm(buff.t(), input2)
81              82
82              83              if bias is not None
                and self.needs_input_grad[3]:
83              -               grad_bias =
                grad_output.sum(0, True)
                84              +               grad_bias =
                grad_output.sum(0,
                keepdim=False)
84              85
85              86              return grad_input1,
                grad_input2, grad_weight,
                grad_bias
```

Comments 0 (0)



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